

Name: _____

Numerical methods

Computer Science BSc

Code: _____

May 9, 2024

Group: _____

Test 2

Solve the below problems! Write your solution of each problem on a separate page.

1. We are using the iteration $x_{k+1} = \frac{\log_2(2x_k + 1)}{16} + 1$ to approximate the solution of the equation $f(x) = \frac{\log_2(2x + 1)}{16} + 1 - x = 0$ in the interval $[\frac{1}{2}, \frac{7}{2}]$.
(2+4+2 points)
 - (a) Show the existence of the root!
 - (b) Prove the convergence of the iteration!
 - (c) Give an error estimation!
2. Consider the equation $f(x) = 2 \cdot \cos(x) - e^{2x} = 0$. (2+2+4 points)
 - (a) Show that the interval $[0, \frac{\pi}{2}]$ contains a root!
 - (b) Write Newton's method to approximate the root!
 - (c) Verify that the conditions of the monotone convergence theorem hold! How to choose the starting point?
3. Interpolate the function $f(x) = \log_2(x) - 2x$ on the base points 1, 2, 4. (4+4+4 points)
 - (a) Give the Newton form of the interpolating polynomial! (No need to expand.)
 - (b) Write the error estimation for the point $x = 3$!
 - (c) What base points should we choose for a quadratic interpolation on the interval $[1, 5]$ to minimize the error bound?
4. Find the best line and parabola fitted on the points (x_i, y_i) given below using least squares approximation. What is the formula for the pseudo inverse of matrices of such overdetermined linear systems? (4+6+2 points)

$$\begin{array}{c|c|c|c|c} x_i & -1 & 0 & 1 & 2 \\ \hline y_i & 0 & 1 & 0 & 2 \end{array}$$

5. Consider the integral $\int_0^2 e^{x^2} dx$. (3+3+4 points)
 - (a) Approximate its value using midpoint, trapezoid and Simpson's formula!
 - (b) Estimate the error of the trapezoid formula!
 - (c) Find m in order to get an approximation of the integral using the composite trapezoid rule T_m within the error 10^{-2} .

Good work!